

ELECTRONIC DESIGN

LAB 2023-2024

Lecture 02

Printed Circuit Boards

Turning schematics into boards

Lab Exercise

due next week_29/02/2024

Design a PCB electronic circuit such as:

Build variable voltage power supply

Build variable current source

Build 555 oscillator

Build closet light timer

Inverting Amplifier, Integrator, Comparator, Oscillator

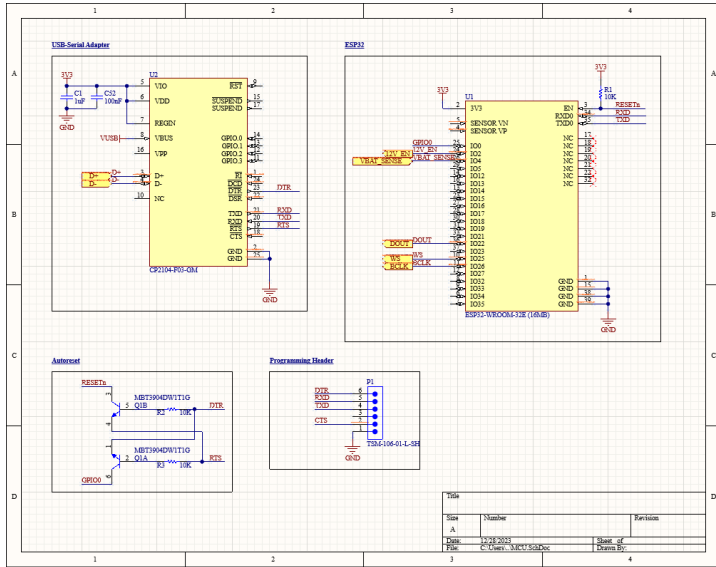
Schmitt Trigger, Monostable Circuit .

Voltage regulator , AC to DC conversion and so on.....

Note: Complete schematic and PCB design on your PC with a summary of your work with your name is required.

How do we layout PCBs?

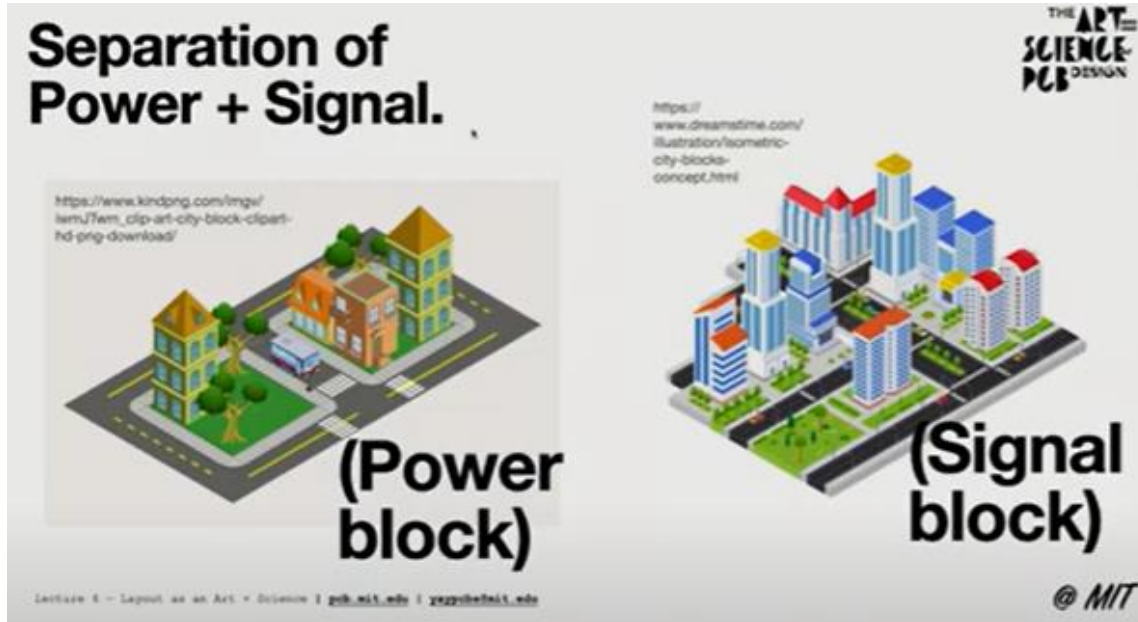
Layout Tools



Given a schematic, how do we convert it into a board that fulfills our requirements?

Routing and Placement Considerations

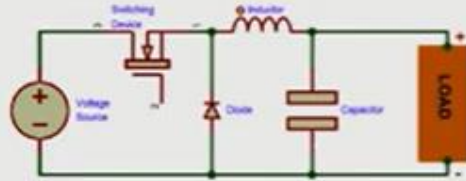
Components on the board, placement and orientation.



Engineering job

Idea is to separate power and signal

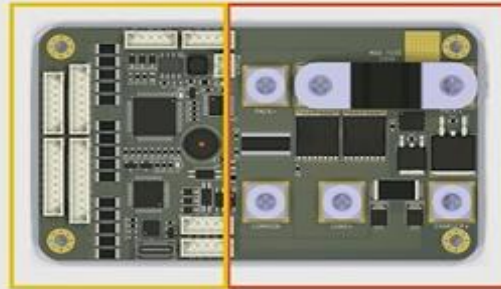
Power, LDOs, Switching Converters, Digital, Analog



<https://components101.com/articles/buck-converter-basics-working-design-and-operation>

(Put these far from sensitive electronics, switching produces noise!)

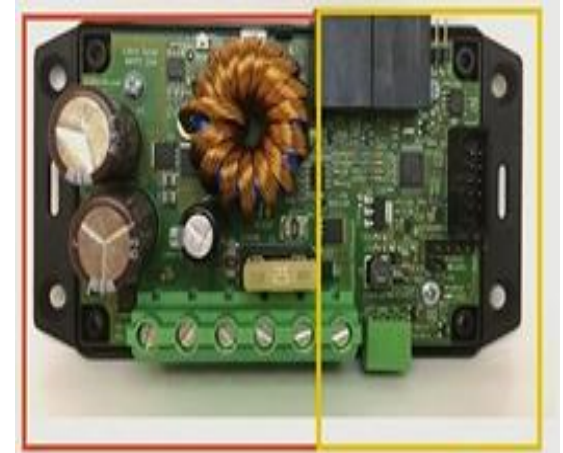
Separation of Power + Signal.



Signal

<https://forum.esk8.news/t/diebiems-fork-ennold-bms-for-6s-to-24s-battery-packs/45490?page=6>

Power



Power

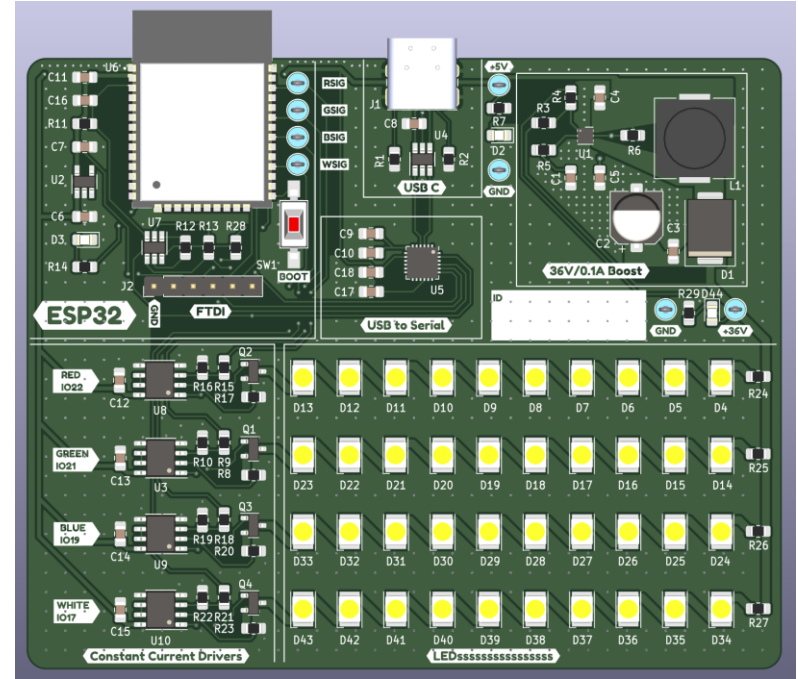
Signal

Routing and Placement Considerations

Separating Circuits

A good practice is to separate the various functional blocks in the schematic into their own physical blocks in layout.

- Each block is typically its own sheet in schematic
- Simplifies reuse of functional blocks
- Easy to follow when debugging
- Conforms with many best practices for minimizing interference between circuits on a board



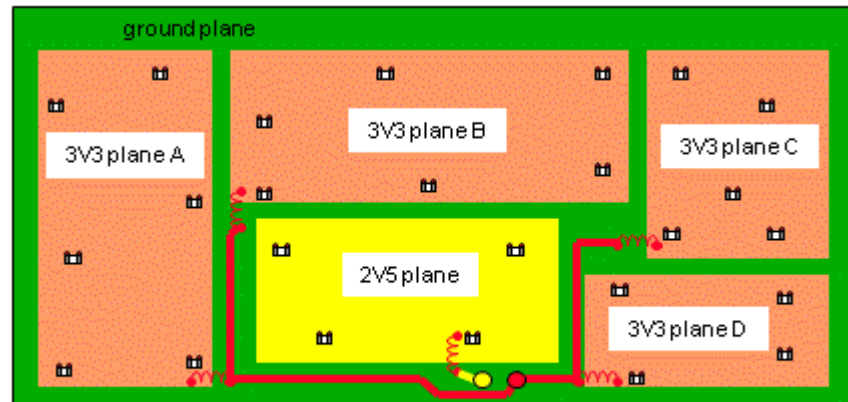
Routing and Placement Considerations

Power Planes

In a multilayer board, parts of or entire internal layers can be assigned to specific power nets to form a power plane

Vias can be used for devices (typically on the top or bottom layers) to easily access a power net

- Reduces number of traces
- Limits power losses
- Isolates noisy power nets



splitting the power plane(s) (not the ground plane) into several segments, each decoupled by a choke, creates "islands" for noisy or sensitive circuits and reduces the impact of plane resonances

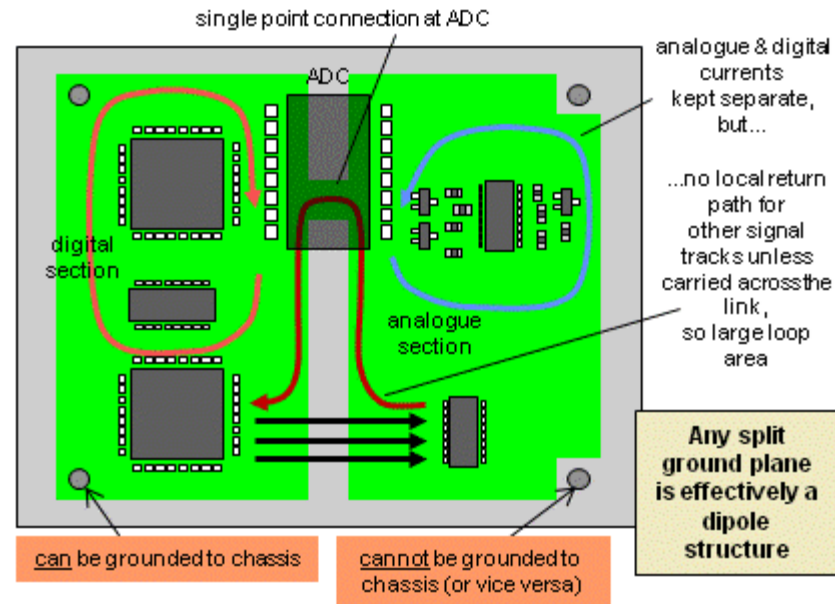
Routing and Placement Considerations

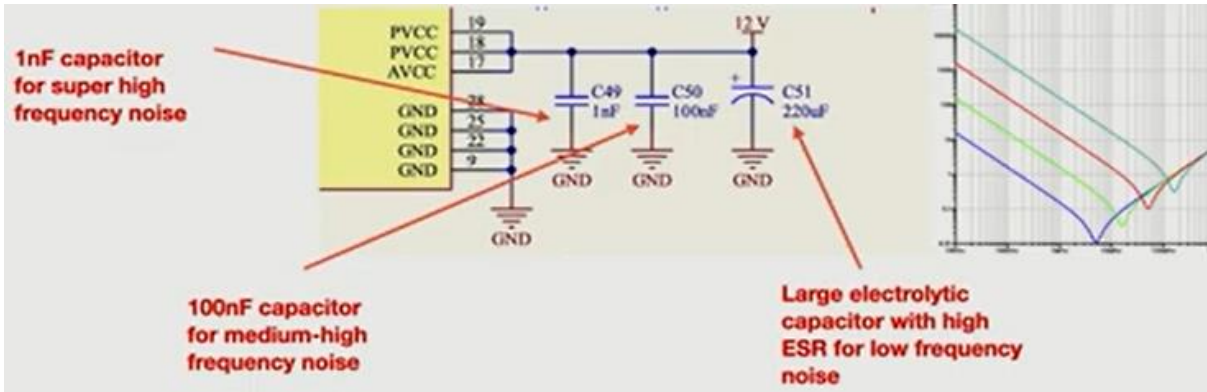
Separate Ground Planes

Ground planes

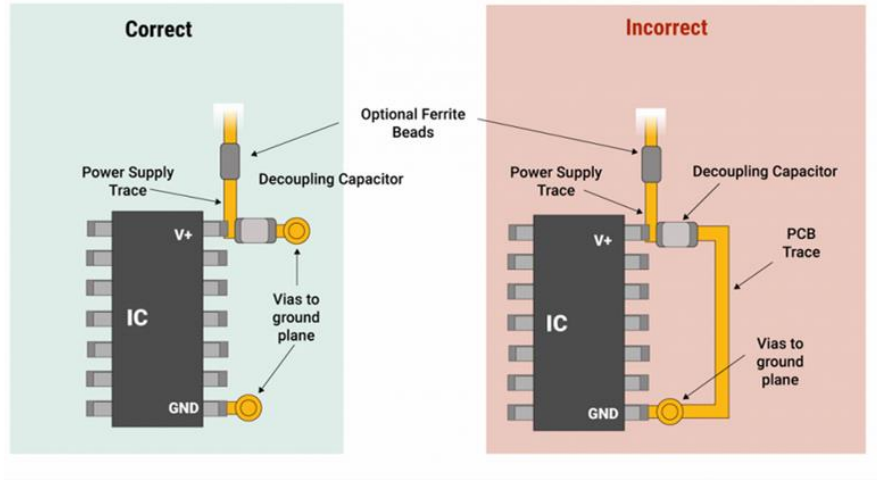
- When high speed digital and high frequency analog circuits are located on the same board

Separate ground planes that are poorly connected to each other as well as careful placement, can isolate noise and prevent each from interfering with the other.





Routing and Placement Considerations



Decoupling capacitors are placed near the power input pins of ICs and provide filtering of noise and transients

Typically, a large- and small-valued capacitor combination is used as close to the IC as possible. (e.g. 100 nF and 200 pF)

Routing and Placement Considerations

Maxwell's Equations

$$1. \quad \nabla \cdot \mathbf{E} = \frac{\rho_v}{\epsilon}$$

Charges produce an electric field

$$2. \quad \nabla \cdot \mathbf{H} = 0$$

Magnetic field lines never end

www.maxwells-equations.com

$$3. \quad \nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t}$$

Changing magnetic field induces electric field

$$4. \quad \nabla \times \mathbf{H} = \epsilon \frac{\partial \mathbf{E}}{\partial t} + \sigma \mathbf{E}$$

Bulk and changing electric field induces magnetic field

Routing and Placement Considerations

Maxwell's Equations

- Charges and fields can induce undesired currents/voltages in a circuit
- As a circuit's operating frequency increases and size decreases, electromagnetic becomes a greater issue
- High currents creates heat in conductors (electron collisions)

Routing and Placement Considerations

Separating Circuits

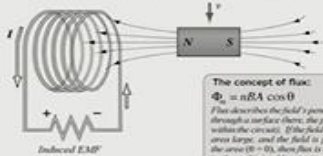
Orientation + Minimization of Trace Length, Inductance Loops.

Parsons • ELECTROMAGNETIC INTERFERENCE • STRELL • Faraday's and Lenz's Laws [179]

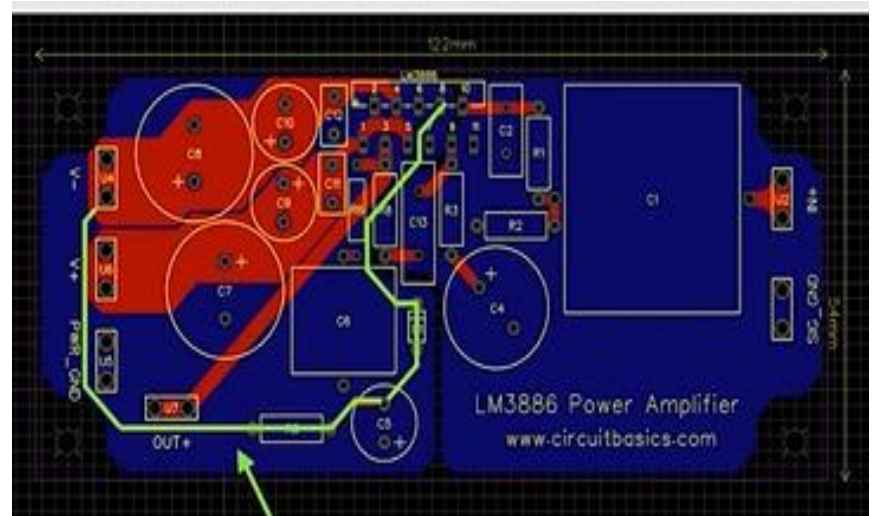
Faraday's Law

$$\mathcal{E} = - \frac{\Delta \Phi_m}{\Delta t}$$

$\mathcal{E}$ = induced emf
 $\frac{\Delta \Phi_m}{\Delta t}$ = rate of change of magnetic flux through the circuit

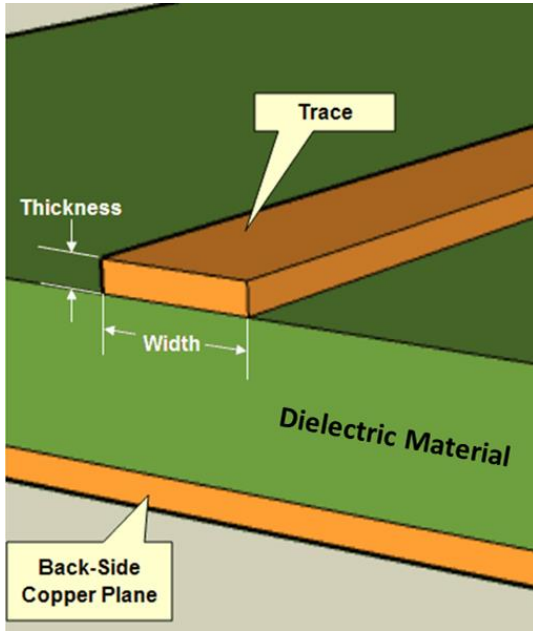


The concept of flux:
 $\Phi_B = nBA \cos \theta$
Flux describes the field's penetration (or flow) through a surface. Here, the plane of the loops within the circuit. If the field is strong and the area large, and the field is perpendicular to the area ($\theta = 0$), then flux is high. Each turn in the coil (n) adds more area.



Calculation trace width and spacing

Trace width and length



- Trace cross section (width and thickness) determines current-carrying capability and resulting temperature rise (keep $< 10^{\circ}\text{C}$ rise)
- Keep traces as short as possible to avoid voltage drops and signal distortions
- For long traces, avoid creating loops to prevent capture of external magnetic flux